TABLE OF CONTENTS

Page

5. WATER QUALITY IMPACT 5-1

5.1 Introduction 5-1
5.2 Environmental Legislation, Standards and Guidelines 5-1
5.3 Description of Environment 5-6
5.4 Water Sensitive Receivers 5-10
5.5 Assessment Methodology 5-12
5.6 Identification, Prediction and Evaluation of Environmental Impacts 5-13
5.7 Mitigation of Adverse Environmental Impacts 5-18
5.8 Cumulative Impacts 5-26
5.9 Evaluation of Residual Impacts 5-27
5.10 Environmental Monitoring and Audit 5-27
5.11 Conclusion 5-27

List of Tables

Table 5.1 Summary of Key Water Quality Objectives for Deep Bay WCZ
Table 5.2 Summary Statistics of Water Quality Monitoring Results
Table 5.3 Summary of Water Quality Monitoring Data for River Indus in 2024
Table 5.4 Summary Statistics of River Water Quality of Ma Tso Lung Stream collected by DSD
Table 5.5 Identified Water Sensitive Receivers

List of Figures

60722948/A10/FIGURE 5.1 Locations of Water Sensitive Receivers and Relevant Water Quality Monitoring Stations

5. WATER QUALITY IMPACT

5.1 Introduction

5.1.1 This section presents an assessment of potential water quality impacts arising from construction and operation of the Project, which has been conducted in accordance with the criteria and guidelines for evaluating and assessing water pollution as stated in Annex 6 and Annex 14 of the "*Technical Memorandum on Environmental Impact Assessment Process*" (EIAO-TM) as well as the requirements given in Clause 3.5.5 and Appendix D of the EIA Study Brief (No. ESB-375/2025) (hereinafter "the Study Brief").

5.2 Environmental Legislation, Standards and Guidelines

Technical Memorandum on Environmental Impact Assessment Ordinance (EIAO-TM)

5.2.1 The EIAO-TM was issued by EPD under Section 16 of the EIAO. Reference sections in the EIAO-TM provide the details of the assessment criteria and guidelines that are relevant to the water quality assessment, including:

- Annex 6 - Criteria for Evaluating Water Pollution
- Annex 14 - Guidelines for Assessment of Water Pollution

Water Pollution Control Ordinance (WPCO)

5.2.2 The WPCO provides the major statutory framework for the protection and control of water quality in Hong Kong. According to the Ordinance and its subsidiary legislation, Hong Kong waters are divided into ten Water Control Zones (WCZs). Corresponding statements of Water Quality Objectives (WQOs) are stipulated for different water regimes (marine waters, inland waters, bathing beaches subzones, secondary contact recreation subzones and fish culture subzones) in the WCZs based on their beneficial uses. According to Item 3.5.5.2 of the Study Brief, the assessment area for this water quality impact assessment covers the Deep Bay WCZ as designated under the WPCO. The WQOs for the watercourses in the Deep Bay WCZ relevant to the water quality impact assessment are listed in **Table 5.1**.

Table 5.1 Summary of Key Water Quality Objectives for Deep Bay WCZ

Parameters	Objectives	Subzone
Aesthetic appearance	Waste discharges shall cause no objectionable odours or discolouration of the water.	Whole Zone
	Tarry residues, floating wood, articles made of glass, plastic, rubber or of any other substances should be absent.	
	Mineral oil should not be visible on the surface. Surfactants should not give rise to a lasting foam.	

Parameters	Objectives	Subzone
	There should be no recognisable sewage-derived debris.	
	Floating, submerged and semi-submerged objects of a size likely to interfere with the free movement of vessels, or cause damage to vessels, should be absent.	
	Waste discharges shall not cause the water to contain substances which settle to form objectionable deposits.	
Bacteria	The level of Escherichia coli should not exceed 610 per 100 mL, calculated as the geometric mean of all samples collected in one calendar year.	Secondary Contact Recreation Subzone and Mariculture Subzone (L.N. 455 of 1991)
	The level of Escherichia coli should be zero per 100 mL, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days.	Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones
	The level of Escherichia coli should not exceed 1000 per 100 mL, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days.	Yuen Long & Kam Tin (Lower) Subzone and other inland waters
	The level of Escherichia coli should not exceed 180 per 100 mL, calculated as the geometric mean of all samples collected from March to October inclusive in one calendar year. Samples should be taken at least 3 times in a calendar month at intervals of between 3 and 14 days.	Yung Long Bathing Beach Subzone (L.N. 455 of 1991)
Colour	Waste discharges shall not cause the colour of water to exceed 30 Hazen units.	Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones
	Waste discharges shall not cause the colour of water to exceed 50 Hazen units.	Yuen Long & Kam Tin (Lower) Subzone and other inland waters

Parameters	Objectives	Subzone
Dissolved Oxygen (DO) within 2 m of the seabed	The concentration of DO should not be less than 2 mg/L for 90% of the sampling occasions during the year.	Outer Marine Subzone excepting Mariculture Subzone
Depth-averaged DO	Waste discharges shall not cause the level of DO to fall below 4 mg/L for 90% of the sampling occasions during the year, values should be taken at 1 metre below surface	Inner Marine Subzone excepting Mariculture Subzone
	Waste discharges shall not cause the level of DO to fall below 4 mg/L for 90% of the sampling occasions during the year, calculated as water column average	Outer Marine Subzone excepting Mariculture Subzone
	The DO level should not be less than 5 mg/L for 90% of the sampling occasions during the year, values should be taken at 1 metre below surface	Mariculture Subzone
	Waste discharges shall not cause the level of DO to be less than 4 mg/L	Yuen Long & Kam Tin (Upper and Lower) Subzones, Beas Subzone, Indus Subzone, Ganges Subzone, Water Gathering Ground Subzones and other inland waters of the Zone
Temperature	Waste discharges should not cause the natural daily temperature range to change by more than 2.0℃.	Whole zone
Salinity	Waste discharges shall not cause the natural ambient salinity level to change by more than 10%.	Whole zone
pH	The pH of the water should be within the range of 6.5-8.5 units. In addition, waste discharges shall not cause the natural pH range to be extended by more than 0.2 units	Marine waters excepting Yung Long Bathing Beach Subzone
	Waste discharges shall not cause the pH of water to exceed the range of 6.5-8.5 units.	Yuen Long & Kam Tin (Upper and Lower) Subzones, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones

Parameters	Objectives	Subzone
	The pH of the water should be within the range of 6.0-9.0 units	Other inland waters
	The pH of the water should be within the range of 6.0-9.0 for 95% of samples. In addition, waste discharges shall not cause the natural pH range to be extended by more than 0.5 units	Yung Long Bathing Beach Subzone
Suspended Solids (SS)	Waste discharges shall neither cause the natural ambient level to be raised by 30% nor give rise to accumulation of SS which may adversely affect aquatic communities.	Marine waters
	Waste discharges shall not cause the annual median of SS to exceed 20 mg/L.	Yuen Long & Kam Tin (Upper and Lower) Subzones, Beas Subzone, Ganges Subzone, Indus Subzone, Water Gathering Ground Subzones and other inland waters
Unionized Ammonia (UIA)	The un-ionized ammoniacal nitrogen level should not be more than 0.021 mg/L, calculated as the annual average (arithmetic mean).	Whole zone
Nutrients	(a) Nutrients shall not be present in quantities sufficient to cause excessive or nuisance growth of algae or other aquatic plants.	Inner and Outer Marine Subzones
	(b) Without limiting the generality of objective (a) above, the level of inorganic nitrogen should not exceed 0.7 mg/L, expressed as annual mean.	Inner Marine Subzones
	(c) Without limiting the generality of objective (a) above, the level of inorganic nitrogen should not exceed 0.5 mg/L, expressed as annual water column average.	Outer Marine Subzones
5-day Biochemical Oxygen Demand (BOD_5)	Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 3 mg/L	Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones

Parameters	Objectives	Subzone
	Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 5 mg/L	Yuen Long & Kam Tin (Lower) Subzone and other inland waters
Chemical Oxygen Demand (COD)	Waste discharges shall not cause the chemical oxygen demand to exceed 15 mg/L	Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones
	Waste discharges shall not cause the chemical oxygen demand to exceed 30 mg/L	Yuen Long & Kam Tin (Lower) Subzone and other inland waters
Toxins	Waste discharges shall not cause the toxins in water to attain such levels as to produce significant toxic, carcinogenic, mutagenic or teratogenic effects in humans, fish or any other aquatic organisms, with due regard to biologically cumulative effects in food chains and to toxicant interactions with each other	Whole zone
	Waste discharges shall not cause a risk to any beneficial uses of the aquatic environment.	Whole zone
Phenol	Phenols shall not be present in such quantities as to produce a specific odour, or in concentration greater than 0.05 mg/L as C_6H_5OH.	Yung Long Bathing Beach Subzone
Turbidity	Waste discharges shall not reduce light transmission substantially from the normal level.	Yung Long Bathing Beach Subzone

Source: Statement of Water Quality Objectives (Deep Bay Water Control Zone).

Technical Memorandum for Effluents Discharged into Drainage and Sewerage Systems, Inland and Coastal Waters (TM-DSS)

5.2.3 Discharge of effluents is subject to control under the WPCO. The TM-DSS gives guidance on the permissible effluent discharges based on the type of receiving waters (foul sewers, storm water drains, inland and coastal waters). The standards control the physical, chemical and microbial quality of effluents. The discharge limits vary with the effluent flow rates and the effluent should comply with the standards for effluent discharged into inland waters for irrigation, pond fish culture or amenity subject to the exact locations. The TM-DSS also specifies the discharge standards into foul sewers leading into the Government's sewage treatment plants. Subject to the flow rate of the effluents, corresponding standards for the effluent discharge into the Government's foul sewers should be followed.

Professional Persons Environmental Consultative Committee Practice Notes (ProPECC PNs)

5.2.4 A ProPECC PN was issued by the EPD to provide guidelines for handling and disposal of construction site discharges. The ProPECC PN 2/24 "*Construction Site Drainage*" provides good practice guidelines for dealing with 10 types of discharge from construction sites. These include surface run-off, groundwater, boring and drilling water, wastewater from concrete batching and/or precast concrete casting, wheel washing water, bentonite slurry, water for testing and/or sterilisation of water retaining structures and water pipes, wastewater from building constructions, acid cleaning, etching and pickling wastewater, and wastewater from site facilities. Practices given in the ProPECC PN 2/24 should be followed as far as possible during construction to minimise the water quality impact due to construction site drainage.

5.2.5 The ProPECC PN 1/23 "*Drainage Plans Subject to Comment by the EPD*" provides guidelines and practices for handling, treatment and disposal of various effluent discharges to stormwater drains and foul sewers during the operation phase. The design of site drainage and disposal of various site effluents generated within the new development area should follow the relevant guidelines and practices as given in the ProPECC PN 1/23.

Environment, Transport and Works Bureau Technical Circular (ETWB TC)

5.2.6 ETWB TC (Works) No. 5/2005 "*Protection of natural streams/rivers from adverse impacts arising from construction works*" provides an administrative framework to better protect all natural streams/rivers from the impacts of construction works. The procedures promulgated under this Circular aim to clarify and strengthen existing measures for protection of natural streams/rivers from government projects and private developments. The guidelines and precautionary mitigation measures given in the ETWB TC (Works) No. 5/2005 should be followed as far as possible to protect the inland watercourse at or near the Project area during construction phase.

Hong Kong Planning Standards and Guidelines (HKPSG)

5.2.7 Chapter 9 of the HKPSG outlines environmental requirements that need to be considered in land use planning. The recommended guidelines, standards and guidance cover the selection of suitable locations for the developments and sensitive uses, provision of environmental facilities, and design, layout, phasing and operational controls to minimise adverse environmental impacts. It also lists out environmental factors that influence land use planning and recommends buffer distances for land uses.

5.3 Description of Environment

5.3.1 According to Item 3.5.5.2 of the EIA Study Brief (No. ESB-375/2025), the assessment area for this water quality impact assessment should include areas within 500 metres from the boundary of the Project and its associated works.

5.3.2 The project boundary falls within the Deep Bay WCZ according to the WPCO. The baseline condition of water bodies in the assessment area was established based on water quality monitoring data collected under the Project "*Development of Lok Ma Chau Loop*" (Environmental Permit no. EP-477/2013) and river water quality monitoring data collected by EPD and DSD. According to water quality monitoring data conducted under the Environmental Monitoring and Audit (EM&A) of EP-

477/2013, water quality monitoring were carried out at the LMC Meander (CS1, IS1 and IS2) and Ping Hang Stream (IS4) within the assessment area. Locations of the monitoring stations are shown in **60722948/A10/FIGURE 5.1**. The water quality monitoring data collected between January 2019 and July 2025 at the monitoring stations are tabulated in **Table 5.2** below.

Table 5.2 Summary Statistics of Water Quality Monitoring Results

Parameter	Unit	Monitoring Stations				WPCO WQO (inland waters)
		LMC Meander			Ping Hang Stream	
		CS1	IS1	IS2	IS4	
Salinity	ppt	2.25 (0.01 - 10.90)	2.20 (<0.01 - 9.80)	2.00 (0.01 - 31.80)	0.10 (<0.01 - 10.40)	Change within ±10% of ambient level
Temperature	℃	27.4 (10.9 - 37.1)	26.6 (9.8 - 36.4)	26.8 (11.1 - 34.8)	25.4 (9.8 - 33.3)	Change within ±2℃
pH	-	7.8 (6.0 - 10.1)	7.7 (5.3 - 10.1)	7.6 (5.9 - 9.9)	7.6 (6.0 - 10.5)	Within 6.0 - 9.0 units
DO	mg/L	7.1 (1.6 - 21.3)	7.3 (1.4 - 19.9)	6.0 (1.9 - 14.9)	4.6 (1.2 - 9.5)	≥ 4 mg/L
DO saturation	%	88.2 (19.5 - 233.7)	91.8 (8.4 - 218.9)	75.7 (6.5 - 190.5)	55.8 (14.7 - 106.8)	Not available
Turbidity	NTU	10.1 (1.1 - 60.4)	12.8 (2.1 - 41.3)	23.5 (0.1 - 999.0)	10.7 (0.1 - 999.0)	Not available
SS	mg/L	15.0 (<2.0 - 127.0)	17.0 (3.0 - 107.0)	28.0 (2.0 - 755.0)	12.0 (<2.0 - >999)	Annual median: ≤ 25 mg/L

Note:
1. Data source: Monthly EM&A Reports for "Development of Lok Ma Chau Loop" under Environmental Permit no. EP-477/2013 (from January 2019 to July 2025).
2. Data presented are medians
3. Data in brackets indicate the ranges.

5.3.3 Based on the water quality monitoring results, the LMC Meander and Ping Hang Stream are freshwater sources, which the water quality would be influenced by Shenzhen River/Deep Bay water. Backflow of turbid water from Shenzhen River to LMC Meander was observed occasionally which affecting the water quality of LMC Meander (especially at IS2, west of the LMC Meander). Stagnant and shallow turbid water was observed at Ping Hang Stream in dry season, causing relatively high SS and turbidity levels were recorded. Apart from high SS and turbidity levels, the measured DO levels at all monitoring stations were high, which indicated that the water quality conditions of these watercourses were generally good.

5.3.4 The nearest EPD's river water quality monitoring station (IN1) is located at Ng Tung River, which is about 5.3 km at upstream of the LMC Meander. A summary of river water quality monitoring data collected in 2024 at EPD station IN1 of Ng Tung River

is presented in **Table 5.3**. According to the EPD's publication "*River Water Quality in Hong Kong in 2024*", the overall WQO compliance rate for Ng Tung River was 86% in 2024, as compared with 32% in 1994. The river monitoring station IN1 situated at Ng Tung River was graded "Good" in Water Quality Index (WQI) grading in 2024.

Table 5.3 Summary of Water Quality Monitoring Data for River Indus in 2024

Parameter	River Indus IN1	WPCO WQO (inland waters)
Dissolved Oxygen (mg/L)	5.6 (3.7 - 6.6)	Indus Subzone: not less than 4 mg/L
pH	6.9 (6.8 - 7.1)	Indus Subzone: within the range of 6.5-8.5
Suspended Solids (mg/L)	12.0 (4.4 – 24.0)	Indus Subzone: annual median not to exceed 20 mg/L
5-day Biochemical Oxygen Demand (BOD_5) (mg/L)	4.2 (1.5 – 9.9)	Indus Subzone: not to exceed 3 mg/L
Chemical Oxygen Demand (mg/L)	16 (5 - 43)	Indus Subzone: not to exceed 15 mg/L
Oil & Grease (mg/L)	<0.5 (<0.5 - <0.5)	Not available
E. coli (count/100mL)	25 364 (3 200 - 190 000)	Not available
Faecal Coliforms (count/100mL)	81 281 (10 000 – 460 000)	Not available
Ammonia-nitrogen (mg/L)	1.050 (0.300 - 1.900)	Not available
Nitrate Nitrogen (NO_3-N) (mg/L)	2.350 (0.460 - 4.900)	Not available
Total Kjeldahl Nitrogen (TKN) (mg/L)	1.75 (0.71 - 2.90)	Not available
Orthophosphate Phosphorus (PO_4) (mg/L)	0.098 (0.052 - 0.200)	Not available
Total Phosphorus (TP) (mg/L)	0.21 (0.11 - 0.31)	Not available
Sulphide (mg/L)	<0.02 (<0.02 - 0.05)	Not available
Aluminium (µg/L)	<50 (<50 - <50)	Not available
Cadmium (µg/L)	<0.1 (<0.1 - 0.3)	Not available
Chromium (µg/L)	<1 (<1 - 2)	Not available

Parameter	River Indus	WPCO WQO (inland waters)
	IN1	
Copper (µg/L)	2 (1 - 3)	Not available
Lead (µg/L)	<1 (<1 - <1)	Not available
Zinc (µg/L)	<10 (<10 - 40)	Not available
Flow (m^3/s)	8.932 (1.078 - 27.797)	Not available

Note:
1. Data source: EPD River Water Quality in Hong Kong in 2024.
2. Data presented are in annual medians of monthly samples; except those for faecal coliforms and E. coli which are in annual geometric means.
3. NM indicates no measurement taken.
4. Figures in brackets are annual ranges.
5. Values at or below laboratory reporting limits are presented as laboratory reporting limits.
6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

5.3.5 In addition, the water quality monitoring data collected by DSD at Ma Tso Lung Stream was also used to establish the baseline water quality conditions. The water quality monitoring data collected between year 2012 and year 2020 at the two monitoring stations are tabulated in **Table 5.4** below.

Table 5.4 Summary Statistics of River Water Quality of Ma Tso Lung Stream collected by DSD

Parameter	Ma Tso Lung		WPCO WQO (inland waters)
	Upstream	Downstream	
Dissolved Oxygen (% saturation)	95 (72 - 103)	78 (53 - 101)	Not available
pH	7.9 (6.6 - 9.1)	7.9 (6.7 - 8.7)	Within 6.5 - 8.5 units
5-day Biochemical Oxygen Demand (BOD_5) (mg/L)	2 (1 - 2)	2 (1 - 4)	Not to exceed 5 mg/L
Ammonia as N (as N mg/L)	0.11 (0.01 - 1.10)	0.18 (0.01 - 0.68)	Not available

Note:
1. Data source: DSD EcoDMS (data collected between year 2012 and year 2020).
2. Data presented are medians
3. Data in brackets indicate the ranges.

5.3.6 Ma Tso Lung Stream is located at Lok Ma Chau. It was a natural stream before 2008 and was channelised in 2010. It flows towards the northwest into Shenzhen River. The majority of drainage catchment of Ma Tso Lung Stream consists of local villages and natural hillsides. Based on the monitoring results, high DO level, low BOD_5 and ammonia-nitrogen levels were recorded which indicated that the water qualities of Ma Tso Lung Stream are excellent with limited contamination by organic and nutrient pollutants.

5.4 Water Sensitive Receivers

5.4.1 Water Sensitive Receivers (WSRs) within the 500m assessment area were identified according to Annex 14 of the EIAO-TM. Representative WSRs were identified based on review of relevant land use plans including Outline Zoning Plans (OZP), information available in the Statutory Planning Portal of the Town Planning Board (TPB) and land status plans published by Lands Department. These include Shenzhen River, LMC meander, Hoo Hok Wai and Ma Tso Lung Nullah. Other potential WSR such as inner Deep Bay and Mai Po Ramsar Site (Mangrove) are located at further downstream of Shenzhen River. However, these areas are considered far away from site (>4km compared to the 500m assessment area) and water quality impact is not anticipated.

5.4.2 Major WSRs within 500m from the Project boundary were identified and are presented in **Table 5.5** and their respective locations are illustrated in **60722948/A10/FIGURE 5.1**.

Table 5.5 Identified Water Sensitive Receivers

ID	Descriptions	Nature	Within Project Area? (Yes/No)	Remarks
S1	Shenzhen River	modified watercourse	No	-
S2	Lok Ma Chau Meander	nature watercourse	Yes (partially)	Section within the Project site will be modified, and its riverbed will be reinstated upon completion of the Project.
S3	Tai Law Hau	natural watercourse	No	-
S4	Ping Hang Stream	nature watercourse	Yes (partially)	Section within the Project site will be temporary diverted during construction.
S5	Tse Koo Hang	nature watercourse	No	-
S6	Ma Tso Lung Nullah /Stream	semi-nature watercourse	Yes (partially)	Viaduct will be provided crossing the downstream channelised section.
S7	near Ma Tso Lung Shun Yee San Tsuen	nature watercourse	No	-
S8	near Liu Pok Road	nature watercourse	No	-
P1	Ponds near Ma Tso Lung Shun Yee San Tsuen	Ponds	No	-
P2	Ponds adjacent to the LMC Meander	Ponds	Yes	Ponds will be isolated and drained during construction. Ponds will be integrated into enhanced wetland upon completion of the Project.

ID	Descriptions	Nature	Within Project Area? (Yes/No)	Remarks
P3	Ponds at Ma Tso Lung, Hoo Hok Wai area and adjacent to the LMC Meander	Ponds	No	-
P4	Pond at Tse Koo Hang	Ponds	No	-
P5	Ponds at Ping Hang	Ponds	No	-
P6	Ponds at Tai Law Hau	Ponds	No	-
EA	Ecological Area (EA)	Compensatory Wetland	Yes (partially)	EA will be isolated during construction. EA will be reinstated upon completion of the Project.
WCA	Wetland Conservation Area	Wetland Conservation Area	Yes (partially)	Partial area within the Project site will be modified.
WBA	Wetland Buffer Area	Wetland Buffer Area	Yes (partially)	Partial area within the Project site will be modified.
CA	Conservation Area	Conservation Area	Yes (partially)	Partial area within the Project site will be modified.

5.4.3 Shenzhen River (S1) locates at the northern portion of assessment area. It is a channelised watercourse which generally flows from northeast to southwest and finally enters the marine water of Deep Bay WCZ. It is the drainage outlet of Ng Tung River, Ping Yuen River and the San Tin streams on Hong Kong side. It has been straightened, widened and deepened to improve its flood prevention level. Tidal influence occurs along the entire Shenzhen River.

5.4.4 LMC Meander (S2) is the largest natural watercourse in the northern portion of assessment area. S2 was originally part of Shenzhen River and was restored to mitigate the loss of fish ponds from training of the Shenzhen River. Water in S2 is upstream of Shenzhen River which originates from adjacent hillside at the southern portion of assessment area and therefore the water quality was less polluted compared to Shenzhen River. Tidal influence was observed occasionally according to the water quality monitoring results as stated in **Section 5.3.3**.

5.4.5 S3 and S4 are small natural watercourses in Tai Law Hau and Ping Hang respectively. These watercourses are upstream tributaries of the LMC Meander (S2), which originate from adjacent hillside at the southern portion of assessment area, flow from south to north and lead northward to S2. As mentioned in **Section 5.3.3**, although stagnant and turbid water was observed at during low water flow condition in dry season, the water quality of S4 is generally good.

5.4.6 S5 is a small natural watercourse in Tse Koo Hang. It originates from adjacent hillside at the southern portion of assessment area, flows from south to north, runs through the farmlands and villages in Tse Koo Hang and finally discharges to the wet grassland/marsh/ponds in Ma Tso Lung.

5.4.7 Ma Tso Lung Nullah/Stream (S6) was a natural stream before 2008 and was trained as a drainage nullah under the project 4156CD - Drainage improvement in Ki Lun Tsuen, Ma Tso Lung, Ying Pun, Shek Tsai Leng and Sha Ling in New Territories. Upstream of S6 is a natural stream near the Ma Tso Lung Shun Yee San Tsuen. As mentioned in **Section 5.3.6**, the water quality of S6 is excellent with limited contamination by organic and nutrient pollutants.

5.4.8 S7 is a small natural watercourse near Ma Tso Lung Shun Yee San Tsuen. This watercourse is upstream tributaries of Ma Tso Lung Nullah/Stream (S6) which collects surface runoff from nearby local villages and nature hillsides, passing through a pond and marsh and finally joins S6.

5.4.9 S8 are several small natural watercourses which collects surface runoff from adjacent hillside (including Crest Hill), flows to Liu Pok at the east and finally discharges to the wet grassland/marsh/ponds in Ta Sha Lok.

5.4.10 A continuous pond area was recorded at Ma Tso Lung, Hoo Hok Wai area and adjacent to Lok Ma Chau Meander (P2 and P3). Majority of the ponds recorded in P2 and P3 are fishponds. Several small ponds (P1, P4, P5, P6) were identified within the assessment area which scattered in the area near Ma Tso Lung Shun Yee San Tsuen, Tse Koo Hang, Ping Hang and Tai Law Hau. These ponds are located near village areas and are inactively managed by nearby villagers, or abandoned.

5.4.11 The Project site is located within the Wetland Conservation Area (WCA) and Wetland Buffer Area (WBA).

5.4.12 Conservation area (CA) is also identified within the assessment area. CA is designated as conservation area under the Approved Lok Ma Chau Loop Outline Zoning Plan (OZP) No. S/LMCL/2, Approved Ma Tso Lung and Hoo Hok Wai OZP No. S/NE-MTL/3 and Approved San Tin Technopole OZP No. S/STT/1 to cover the LMC Meander and the wetland and fish ponds which form an integral part of the wetland ecosystem at Hoo Hok Wai and Deep Bay area.

5.4.13 Ecological Area (EA) is identified within the assessment area. EA was created along the southeastern edge of the Loop, covers an area of approximately 12.78 ha to compensate for the loss of wetland habitats due to the Loop's development (register no.: AEIAR-176/2013).

5.5 Assessment Methodology

5.5.1 The Project falls within the Deep Bay WCZ as designated under the WPCO (Cap. 358). The assessment area should include areas within 500m from the boundary of the Project and its associated works as identified in the EIA Study. The WSRs in the vicinity of the Project have been identified. Potential sources of water quality impact that may arise during the construction and operation stages of the Project were described, including point discharges and non-point sources to surface water runoff, sewage from workforce and polluted discharge generated from the Project. All the identified sources of potential water quality impact at the identified WSRs were evaluated and their impact significance determined with reference to the WQOs or criteria covered in Annex 6 and 14 of the EIAO-TM. Practical water pollution control measures were recommended to mitigate identified water quality impacts.

5.6 Identification, Prediction and Evaluation of Environmental Impacts

Construction Phase

5.6.1 General construction works for the Project would be land-based only and within the Project site. The potential sources of water quality impact associated with the land-based works would include:

- Construction works in LMC Meander and EA;
- Wastewater from general construction activities;
- Construction site run-off;
- Construction works near watercourses;
- Diversion of watercourse;
- Removal of ponds and wet areas;
- Accidental spillage of chemicals;
- Sewage from construction workforce; and
- Groundwater infiltration arising from formation of underpass/depressed road.

Construction Works in LMC Meander and EA

5.6.2 Underpass road will be constructed across the LMC Meander and EA. Excavation works will be required during cut-and-cover underpass road construction in the LMC Meander and EA, which may lead to elevated suspended solids (SS) levels and dispersion of SS to the downstream Shenzhen River/Deep Bay waters if in contact with waters in the Meander and EA.

5.6.3 To separate the construction works from the meander and EA, cofferdams will be used to enclose the excavation works. The construction of underpass road across the LMC Meander will be divided into two stages (half width of the Meander per each stage). Details on construction of cut-and-cover underpass enclosed by cofferdams are discussed in **Section 2**. Water pumps will be used to collect any construction site runoff and ingress/seepage water within the cofferdams. The collected construction site runoff and ingress/seepage water will be diverted to the on-site wastewater treatment facilities for treatment to satisfactory levels before discharge. Silt curtains shall be deployed to completely enclose the cofferdams prior to setting up excavation works, and installation of cofferdams. Silt curtains shall only be removed after completion of cut-and-cover construction works, and removal of cofferdams. The riverbed of the LMC Meander (section for underpass road) and EA will be reinstated upon completion of the Project.

5.6.4 Excavated materials (including sediment) may be generated from the construction works in the LMC Meander and EA. Mitigation measures for handling and disposal of excavated materials and sediment as recommended in **Section 5.7.25** should be followed to minimise the potential environmental impacts arising from the excavated materials.

5.6.5 The construction of underpass road across the LMC Meander will be conducted by phases and only one cofferdam will be installed in each stage. Since the cross-section area of the LMC Meander to be blocked by the cofferdams is anticipated to be localised and temporary, the change in flow regime would be localised and any local water quality impact would expect to be transient. As a result, the potential impact on water quality and hydrology of the LMC Meander and EA would be insignificant, no adverse water quality would be expected if the mitigation measures detailed in **Sections 5.7.1** to **5.7.4** are implemented properly.

Construction Site Runoff and Wastewater from General Construction Activities

5.6.6 Wastewater generated from construction activities, including general cleaning and polishing, wheel washing, dust suppression and utility installation may contain high SS concentrations and a certain amount of grease and oil. Wastewater may also be generated from the accumulation of solid waste such as debris, rubbish, plastic package and construction materials. Sediment laden run-off during site formation works, if uncontrolled, may carry pollutants (absorbed onto the particle surfaces) into the nearby water bodies. Potential pollution sources of site run-off may include:

- Run-off and erosion of exposed bare soil and earth, drainage channel, earth working area and stockpiles;
- Release of any bentonite slurries, concrete washings and other grouting materials with construction run-off or storm water;
- Wash water from dust suppression sprays and wheel washing facilities; and
- Fuel, oil and lubricants from maintenance of construction vehicles and equipment.

5.6.7 During rainstorms, site run-off would wash away the soil particles on unpaved lands and areas with the topsoil exposed. The run-off is generally characterised by high concentration of SS. Release of uncontrolled site run-off would increase the SS levels and turbidity in the nearby water environment. Site run-off may also wash away soil particles that were contaminated by the construction activities and therefore cause water pollution.

5.6.8 Wind-blown dust would be generated from exposed soil surfaces in works areas. It is possible that wind-blown dust would fall directly onto the nearby water bodies when a strong wind occurs. Dispersion of dust within the works areas may increase the SS levels in surface run-off causing a potential impact to the nearby sensitive receivers.

5.6.9 It is important that proper site practice and good site management be followed to prevent run-off with high level of SS from entering the surrounding waters. Best Management Practices (BMPs) in controlling construction site discharges are recommended for this Project as described in **Sections 5.7.5** to **5.7.18**. In view of the sensitive nature of the Deep Bay WCZ, an Emergency Response Plan (ERP) should also be developed to minimise the potential impact from construction site discharges under failure of treatment facilities during emergency situations or inclement weather. An outline of the ERP is provided in **Sections 5.7.34** to **5.7.35**. With the implementation of BMPs to control construction site effluents and ERP to minimise water quality impact under emergency situations, disturbance of water bodies would be avoided and deterioration in water quality would be minimal.

Construction Works near Watercourses

5.6.10 Watercourses were identified within the Site as identified in **Section 5.4**. Construction works in close proximity to the inland watercourses may pollute the inland waters due to potential release of construction wastes and wastewater. Construction wastes and wastewater are generally characterised by high concentration of SS and elevated pH. Adoption of good housekeeping and mitigation measures would reduce the generation of construction wastes and potential water pollution. The implementation of measures to control run-off and drainage water will be important for the construction works adjacent to the inland water in order to prevent run-off and drainage water with high levels of SS from entering the water environment. With the implementation of BMPs as described in ProPECC PN 2/24 "*Construction Site Drainage*" and provision of mitigation measures as specified in ETWB TC (Works) No. 5/2005 "*Protection of natural streams / rivers from adverse impacts arising from construction works*" (also see **Sections 5.7.5** to **5.7.19**), it is anticipated that water quality impacts would be minimised.

Diversion of Watercourse

5.6.11 Temporary diversion will be required during cut-and-cover construction of depressed road across Ping Hang Stream (S4). Diversion of watercourse would involve excavation and construction works for diverting the water flow from the existing route to the temporary drainage system. Potential impacts may be generated by release of construction materials, wastewater, excavated sediment, spillage and contaminants to the receiving waters in the downstream (due to soil excavation for construction of temporary drainage). All these construction works should be undertaken in dry conditions to avoid potential water quality impacts upon the downstream water quality.

5.6.12 The tentative works sequence for diversion of watercourse should be duly planned to minimise water quality impacts as described below. A temporary drainage at most downstream should first be established to intercept and divert the upstream flow. The temporary drainage should be constructed and completed before dewatering of the existing watercourse. Construction of the temporary drainage should be undertaken in a dry condition prior to receiving any water flow. Precautionary measures in Appendix D of ETWB TC No. 5/2005, such as temporary isolation to other connected watercourse using sandbags and silt curtains, should be used to isolate the works areas from neighbouring waters. Dewatering of watercourse should be performed by diverting the water flow to temporary drainage. Silt removal facilities should be adopted to treat the wastewater from dewatering operations prior to discharge. Construction works at watercourse should be undertaken only after dewatering operation is fully completed to avoid water flow in the works area. After completion of the construction works, the works area shall be cleaned up before receiving any water flow or connecting to any existing watercourse.

5.6.13 Excavated materials (including sediment) may be generated from the diversion of watercourse. Mitigation measures for handling and disposal of excavated materials and sediment as recommended in **Section 5.7.25** should be followed to minimise the potential water quality impacts arising from the excavated materials.

5.6.14 Mitigation measures for protection of downstream water quality from diversion of watercourses are described in **Sections 5.7.20** to **5.7.23**. With adoption of the recommended mitigation measures, no adverse water quality impacts are anticipated.

Removal of Ponds and Wet Areas

5.6.15 Three ponds (P2) will be drained during construction and integrated into enhanced wetland after completion of works. Partial area of wet grassland (in WBA) in between the Tse Koo Hang Stream (S5) and WCA will also be removed under this Project.

5.6.16 All the affected ponds and wet areas should be isolated and not connected to any existing watercourse. The associated construction works would include draining the water in ponds and wet areas before filling up these areas or before commencement of any excavation and construction works. The water of these ponds and wet areas to be drained would probably be sediment-laden and would carry a certain level of pollutants. Direct dumping of these drained waters to the nearby watercourse will not be allowed.

5.6.17 The drained water generated from dewatering of the ponds/wet areas to be removed should be temporarily stored in appropriate storage tanks or containers for reuse on-site as far as practicable and any surplus water should be tankered away for disposal at the Sewage Treatment Works (STW). Any surplus water should be treated as necessary before disposal to the STW in compliance with the TM-DSS. In order to minimise the potential impact, dewatering works at ponds and wet areas should be conducted within dry season to minimise the quantity of drained water. No direct discharge of drained water from these construction works will be allowed.

5.6.18 Excavated materials (including sediment) may be generated from the construction works in ponds and wet areas. Mitigation measures for handling and disposal of excavated materials and sediment as recommended in **Section 5.7.25** should be followed to minimise the potential water quality impacts arising from the excavated material. Together with the adoption of the mitigation measures for removal of ponds and wet areas as recommended in **Sections 5.7.24**, no adverse water quality impact would be expected.

Accidental Spillage of Chemicals

5.6.19 The use of engine oil and lubricants and their storage as waste materials have the potential to create impacts on the water quality if spillage occurs. Waste oil may infiltrate into the surface soil layer, or runoff into adjacent waterbodies, increasing hydrocarbon levels. Groundwater pollution may also rise from the improper use and storage of chemical and petroleum products within the site area where groundwater infiltrates into the area. Infiltration of groundwater may occur at areas where there are faults and/or fissures in the rock mass. The potential impacts could however be mitigated by practical mitigation measures and good site practices (as given in **Sections 5.7.26** to **5.7.28**).

Sewage from Construction Workforce

5.6.20 During the construction of the Project, the workforce on site will generate sewage effluents, which are characterised by high level of BOD, ammonia and *E. coli*. According to Table T-2 of "*Guidelines for Estimating Sewage Flows for Sewage Infrastructure Planning*" (GESF), the unit flow factor (UFF) is 0.23 m^3/day/employee (0.08 m^3/day/employee for commercial employee and 0.15 m^3/day/employee for employees of construction activities). For every 100 construction workers working simultaneously at the construction site, about 23 m^3 of sewage would be generated per day. Potential water quality impacts upon the local drainage and fresh water system may arise from these sewage effluent, if uncontrolled.

5.6.21 However, this temporary sewage can be adequately handled by interim sanitary facilities, such as portable chemical toilets. According to Section 2.2 of "*Reference Materials on Construction Site Facilities for Workers*" that issued by the Construction Industry Council (CIC), the number of toilet facilities provided on site shall be at a ratio of not less than 1 for every 30 workers. The number of the chemical toilets required for the construction site should be subject to later detailed design, the capacity of the chemical toilets, and contractor's site practices. A licensed contractor should be employed to provide appropriate and adequate portable toilets and be responsible for appropriate disposal and maintenance.

5.6.22 Provided that sewage is not discharged directly into stormwater drains or inland waters adjacent to the construction site, and temporary sanitary facilities are used and properly maintained, no adverse water quality impact would be anticipated. Mitigation measures and good site practices as given in **Sections 5.7.29** to **5.7.30** should be implemented.

Groundwater Infiltration arising from Formation of Underpass/Depressed Road

5.6.23 The underpass / depressed road will be constructed by cut-&-cover method which may result in infiltration of groundwater. The major concern of groundwater infiltration from the underpass/depressed road construction activities would be the increase in site runoff and the associated potential drawdown of groundwater in any soil and aquifer layers. Infiltrated water would carry away silt from site into the site drainage and any potential drawdown of groundwater would result in different degrees of settlement and dewatering of surface water features. Consideration should be taken in the underground structure design to minimise the infiltration of groundwater. With the implementation of the recommended groundwater control/mitigation measures as described in **Section 5.7.31** to **5.7.33**, the potential groundwater infiltration would be minimised and no adverse water quality impact would be anticipated.

Operation Phase

5.6.24 Potential water quality impacts associated with the operation phase would include:

- Non-point source surface runoff from new impervious areas.

Surface Runoff

5.6.25 Surface runoff to be generated from the Project site is known as non-point source pollution. Any increase in paved area within the Project site will increase the chance of surface runoff. The presence of oil, grease and grit on surfaces of road within the Project site could be washed into the nearby drainage system or even into the watercourses during rainfall event. Surface runoff generated from the paved or developed areas may contain debris, refuse and dust. Practices that cleaning agents used for road washing may also affect the water quality of the nearby receiving water.

5.6.26 The Project was expected to have an additional 6.78 ha paved area when in operation. The EPD study "*Update on Cumulative Water Quality and Hydrological Effect of Coastal Developments and upgrading of Assessment Tool*" suggested that only rainfall events of sufficient intensity and volume would give rise to runoff. It is assumed that only a daily rainfall of greater than 10 mm per day and a rainfall intensity greater than 2 mm/hr would give rise to runoff. The runoff percentage is based on the average rainfall data from the Hong Kong Observatory. The daily runoff value and the runoff percentage are calculated as follows:

Daily runoff value (mm/day) = Averaged daily rainfall (mm/day) x Runoff percentage

where: Runoff percentage = (Sum of rainfall for days with total rainfall > 10 mm and with maximum rainfall intensity > 2 mm/hr) / Total rainfall of the month × 100%

5.6.27 The rainfall data obtained from the Hong Kong Observatory (HKO) in the period from 2019 to 2023 were analysed to estimate the average daily runoff value (mm/day) and runoff percentage (%) in each month over the year. Based on the past 5-year rainfall data (between 2019 and 2023), the average annual rainfall and runoff percentage were calculated to be 2,442 mm/year and 80.9% respectively. Therefore, it was assumed that only 1,975 mm of the 2,442 mm annual rainfall would be considered as effective rainfall that would generate surface runoff. With consideration of the paved area and a runoff coefficient of 1 for paved surface, the overall daily runoff generated from the Project was estimated to be about 367 m^3/day (i.e.1,975/1,000/365*67,800*1).

5.6.28 To minimise the impact from road runoff, all the road works planned under the Project should be designed with adequate drainage system and appropriate oil interceptors, as required. The design of stormwater drains shall follow the relevant guidelines and practices as given in the ProPECC PN 1/23 "*Drainage Plans subject to Comment by the EPD*". With implementation of recommended measures as described in **Sections 5.7.37** to **5.7.45**, the potential water quality impact due to surface runoff is expected to be minor and no adverse water quality impact from non-point source surface run-off is expected.

5.7 Mitigation of Adverse Environmental Impacts

Construction Phase

Construction works in LMC Meander and EA

5.7.1 The method and sequence for construction of underpass road across the LMC Meander and EA should be carefully designed so that wastewater and sediment laden water generated from the excavation works would be confined and physically separated from the watercourse. The riverbed of the LMC Meander (section for underpass road) and EA should be reinstated upon completion of the Project. For reinstatement of the EA, a reinstatement and monitoring plan for the EA should be prepared by the Contractor and submitted to AFCD and EPD for agreement prior to commencement of construction works (also see **Section 8**).

5.7.2 For construction works conducted in the LMC Meander and EA, cofferdams should be used to enclose the excavation works. The construction of underpass road across the LMC Meander should be conducted by phases and only one cofferdam would be installed in each stage. The detail design of the cofferdams shall be conducted by the Contractor during the construction phase to fulfil the requirements in DSD Technical Circular No. 1/2017 "*Temporary Flow Diversions and Temporary Works Affecting Capacity in Stormwater System*" for DSD approval in order to formulate feasible options of these temporary structure.

5.7.3 Water pumps should be used to collect any construction site runoff and ingress/seepage water within the cofferdams. The collected construction site surface runoff and ingress/seepage water should be diverted to the on-site wastewater treatment facilities for treatment to satisfactory levels before discharged. There is a need to apply to EPD for a discharge licence for discharging effluent from the

construction site under the WPCO. The discharge quality must meet the requirements specified in the discharge licence and follow the TM-DSS.

5.7.4 To further minimise any adverse water quality impact during the excavation works in the LMC Meander and EA, silt curtains should be deployed to completely enclose the cofferdams prior to setting up excavation works, and installation of cofferdams. Silt curtains should only be removed after completion of cut-and-cover construction works, and removal of cofferdams. The Contractor should be responsible for the design, installation and maintenance of the silt curtain to minimise the impacts on water quality. The design and specification of the silt curtains should be submitted by the Contractor to the Project Manage for approval.

Construction Site Runoff and General Construction Activities

5.7.5 Control of potential pollution of nearby water bodies during the construction phase of the Project should be achieved by measures to:

- prevent or minimise the likelihood of pollutants (generated from construction activities) being in contact with rainfall or run-off; and
- abate pollutants in the stormwater surface run-off prior to the discharge of surface run-off to the nearby water bodies.

5.7.6 These principle objectives should be achieved by implementation of the Best Management Practices (BMPs) of mitigation measures in controlling water pollution. The guidelines for handling and disposal of construction site discharges as detailed in the ProPECC PN 2/24 "*Construction Site Drainage*" should be followed, where applicable. All effluent discharged from the construction site should comply with the standards stipulated in the discharge licence under the WPCO. The following measures are recommended to protect water quality of the inland and coastal waters, and when properly implemented should be sufficient to adequately control site discharges so as to avoid water quality impacts.

Construction Site Runoff

5.7.7 Surface run-off from construction site should be discharged into storm water drains via adequately designed on-site wastewater treatment facility such as sand traps, silt traps, sedimentation tanks and sediment basins. Channels, earth bunds or sand bag barriers should be provided on site to properly direct surface run-off to such silt removal facilities. Perimeter channels at site boundaries should be provided where necessary to intercept surface run-off from outside the site so that it will not wash across the site. Catchpits and perimeter channels should be constructed in advance of site formation works and earthworks.

5.7.8 Silt removal facilities, channels and manholes should be maintained and the deposited silt and grit should be removed regularly, at the onset of and after rainstorm to ensure that these facilities are functioning properly at all times. Before disposal at the public fill reception facilities, the deposited silt and grit should be solicited in such a way that it can be contained and delivered by dump truck instead of tanker truck. Any practical options for the diversion and realignment of drainage should comply with both engineering and environmental requirements in order to provide adequate hydraulic capacity of all drains.

5.7.9 Construction works should be programmed to minimise soil excavation in the rainy seasons (generally from April to September). If soil excavation cannot be avoided in these months or at any time of year when rainstorms are likely, for the purpose of preventing soil erosion, temporarily exposed slope surfaces should be covered (e.g. by tarpaulin), and temporary access roads should be protected by crushed stone or gravel, as excavation proceeds. Intercepting channels should be provided (e.g. along the crest / edge of excavation) to prevent surface run-off from washing across exposed soil surfaces. Arrangements should always be in place to ensure that adequate surface protection measures can be safely carried out well before the arrival of rainstorm.

5.7.10 Earthworks final surfaces should be well compacted and the subsequent permanent work or surface protection works should be carried out immediately after the final surfaces are formed to prevent erosion caused by rainstorms. Appropriate drainage like intercepting channels should be provided where necessary.

5.7.11 Measures should be taken to minimise the ingress of rainwater into trenches. If excavation of trenches in the rainy season is necessary, they should be dug and backfilled in short sections of length. Rainwater pumped out from trenches or foundation excavations should be discharged into storm drains via silt removal facilities.

5.7.12 Open stockpiles of construction materials (e.g. aggregates, sand and fill material) on sites should be covered with tarpaulin or similar fabric during rainstorms. Measures should be taken to prevent the washing away construction materials, soil, silt or debris into any drainage system.

5.7.13 Manholes (including newly constructed ones) should always be adequately covered and temporarily sealed so as to prevent silt, construction materials or debris from getting into the drainage system, and to prevent storm run-off from getting into foul sewers. Discharge of surface run-off into foul sewers must always be prevented in order not to unduly overload the foul sewerage system.

Boring and Drilling Water

5.7.14 Water used in ground boring and drilling for site investigation or rock/soil anchoring should as far as practicable be re-circulated after sedimentation. When there is a need for final disposal, the wastewater should be discharged into storm drains via silt removal facilities.

Wheel Washing Water

5.7.15 All vehicles and plants should be cleaned before they leave a construction site to ensure no earth, mud, debris and the like is deposited by them on roads. A wheel washing bay should be provided at every site exit if practicable and wash-water should have sand and silt settled out or removed before discharging into storm drains. The section of construction road between the wheel washing bay and the public road should be paved to reduce vehicle tracking of soil and to prevent site run-off from entering public road drains.

Acid Cleaning, Etching and Pickling Wastewater

5.7.16 Acidic wastewater generated from acid cleaning, etching, pickling and similar activities should be neutralised to within the pH range of 6 to 10 before discharging

into foul sewers. If there is no public foul sewer in the vicinity, the neutralised wastewater should be tankered off site for disposal into foul sewers or treated to a standard acceptable for discharge into storm water drains and the receiving waters.

Rubbish and Litter

5.7.17 Good site practices should be adopted to remove rubbish and litter from construction site so as to prevent the rubbish and litter from spreading from the site area. The construction site should be cleaned on a regular basis.

Effluent Discharge

5.7.18 There is a need to apply to EPD for a discharge licence for discharge of effluent from the construction site under the WPCO. The discharge quality must meet the requirements specified in the discharge licence. All the runoff and wastewater generated from the works areas should be treated so that it satisfies all the standards listed in the discharge licence under the WPCO. The beneficial uses of the treated effluent for other on-site activities such as dust suppression, wheel washing and general cleaning etc., can minimise water consumption and reduce the effluent discharge volume. If monitoring of the treated effluent quality from the works areas is required during the construction phase of the Project, the monitoring should be carried out in accordance with the relevant WPCO licence.

Construction Works near Watercourses

5.7.19 To minimise the potential water quality impacts from the construction works located near any inland watercourses, the practices outlined in ETWB TC (Works) No. 5/2005 "*Protection of natural streams / rivers from adverse impacts arising from construction works*" should be adopted where applicable. Relevant mitigation measures are listed below:

- Impermeable sheet piles and / or cofferdams should be used as required to divert water flow from the construction works area so that all the construction works would be undertaken within a dry zone and physically separated from the watercourses.
- Construction works close to the inland watercourses should be carried out in dry season as far as practicable where the flow in the surface channel/stream is low.
- The use of less or smaller construction plants may be specified in works areas close to the inland watercourses.
- Temporary storage of materials (e.g. equipment, filling materials, chemicals and fuel) and temporary stockpile of construction materials should be located well away from any watercourses during carrying out of the construction works.
- Stockpiling of construction materials and dusty materials should be covered and located away from the any watercourses.
- Construction debris and spoil should be covered up and/or disposed of as soon as possible to avoid being washed into the nearby water receivers.

- Construction activities, which generate large amount of wastewater, should be carried out at a distance away from the watercourses, where practicable.

- Mitigation measures to control site run-off from entering the nearby water environment should be implemented to minimise water quality impacts. Surface channels should be provided along the edge of the waterfront within the work sites to intercept the run-off.

- Construction effluent, site run-off and sewage should be properly collected and/or treated.

- Any temporary works site inside the stormwater watercourses should be temporarily isolated, such as by placing of sandbags or silt curtains with lead edge at bottom and properly supported props to prevent adverse impact on the stormwater quality.

- Proper shoring may need to be erected in order to prevent soil/mud from slipping into the inland watercourses.

Diversion of Watercourse

5.7.20 The construction method and sequence of diversion of watercourse should be carefully designed so that all the construction works including any excavation would be undertaken within a dry zone and physically separated from the watercourses downstream.

5.7.21 The tentative works sequence for diversion of watercourse should be duly planned to minimise water quality impacts as described below. A temporary drainage at most downstream should first be established to intercept and divert the upstream flow. The temporary drainage should be constructed and completed before dewatering of the existing watercourse. Construction of the temporary drainage should be undertaken in a dry condition prior to receiving any water flow. Precautionary measures in Appendix D of ETWB TC No. 5/2005, such as temporary isolation to other connected watercourse using sandbags and silt curtains, should be used to isolate the works areas from neighbouring waters. Dewatering of watercourse should be performed by diverting the water flow to temporary drainage. Silt removal facilities should be adopted to treat the wastewater from dewatering operations prior to discharge. Construction works at watercourse should be undertaken only after dewatering operation is fully completed to avoid water flow in the works area. After completion of the construction works, the works area shall be cleaned up before receiving any water flow or connecting to any existing watercourse.

5.7.22 The Contractor should provide a dry zone for all the construction works to be undertaken in watercourses and stormwater drainage following the tentative works sequence as described above or using other approved methods as appropriate to suit the works condition. The flow diversion works should be conducted in dry season, where possible, when the flow in the watercourse is low. The wastewater and ingress water from the site should be properly treated to comply with the requirements of the WPCO licence and the TM-DSS before discharge.

5.7.23 The site practices outlined in the ProPECC PN 2/24 "*Construction Site Drainage*" and ETWB TC (Works) No. 5/2005 "*Protection of natural streams/rivers from adverse impacts arising from construction works*" as recommended in **Sections 5.7.5** to

5.7.19 should also be followed to minimise the potential water quality impacts from construction in watercourses.

Removal of Ponds and Wet Areas

5.7.24 Construction works at the existing affected ponds/wet areas should be conducted only after dewatering of these ponds/wet areas is fully completed. The drained water generated from the dewatering of these ponds/wet areas to be removed should be temporarily stored in appropriate storage tanks or containers for reuse on-site as far as possible. Any surplus drained water should be tankered away for disposal at the STW in a controlled manner. No direct discharge of drained water to the stormwater drainage system or marine water should be allowed.

Disposal of Excavated Materials and Sediment

5.7.25 All excavated materials generated from construction of underpass road across the LMC Meander, diversion of watercourses, partial removal and construction works in ponds and wet areas should be collected and handled in compliance with the Waste Disposal Ordinance. Excavated sediment, if any, generated from the excavation activities in the channels should be tested and classified in accordance with the Section 4.2.1 of Project Administration Handbook (PAH) for Civil Engineering Works "*Marine Management of Dredged/Excavated Sediment*" for determining the disposal arrangement for the sediment. The disposal of excavated sediments should be minimised according to the relevant requirements in **Section 6**. No direct disposal of the construction wastes or excavated materials into the stormwater drainage system and marine water would be allowed.

Accidental Spillage of Chemicals

5.7.26 Contractor must register as a chemical waste producer if chemical wastes would be produced from the construction activities. The Waste Disposal Ordinance (Cap 354) and its subsidiary regulations in particular the Waste Disposal (Chemical Waste) (General) Regulation, should be observed and complied with for control of chemical wastes. The Contractor is also recommended to develop management procedures for chemicals used and prepare an emergency spillage handling procedure to deal with chemical spillage in case of accident occurs.

5.7.27 Any service shop and maintenance facilities should be located on hard standings within a bunded area, and sumps and oil interceptors should be provided. Maintenance of vehicles and equipment involving activities with potential for leakage and spillage should only be undertaken within the areas appropriately equipped to control these discharges.

5.7.28 Disposal of chemical wastes should be carried out in compliance with the Waste Disposal Ordinance. The Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes published under the Waste Disposal Ordinance details the requirements to deal with chemical wastes. General requirements are given as follows:

- Suitable containers should be used to hold the chemical wastes to avoid leakage or spillage during storage, handling and transport.
- Chemical waste containers should be suitably labelled, to notify and warn the personnel who are handling the wastes, to avoid accidents.

- Storage area should be selected at a safe location on site and adequate space should be allocated to the storage area.

Sewage from Construction Workforce

5.7.29 No discharge of sewage to the stormwater drains or inland water will be allowed. Adequate and sufficient portable chemical toilets should be provided in the works areas to handle sewage from construction workforce. A licensed collector should be employed to clean and maintain the chemical toilets on a regular basis.

5.7.30 Notices should be posted at conspicuous locations to remind the workers not to discharge any sewage or wastewater into the surrounding environment. Regular environmental audit of the construction site should be conducted to provide an effective control of any malpractices and achieve continual improvement of environmental performance on site.

Groundwater Infiltration arising from Formation of Underpass/Depressed Road

5.7.31 The underpass/depressed road will be constructed by cut-&-cover method and fully enclosed by cofferdams to minimise the intrusion of groundwater during excavation. Soil excavation will be commenced after installed the pile walls of cofferdams. The intrusion of groundwater through cofferdams and/or watertight structures during soil excavation is therefore considered insignificant.

5.7.32 In case seepage of groundwater occurs, groundwater should be pumped out from works areas and discharged to the drainage system via silt trap. Groundwater from dewatering process should also be discharged to the storm system via silt removal facilities.

5.7.33 In the event of excessive groundwater infiltration being observed during underpass / depressed road construction, grouting should be applied as far as practicable to limit the groundwater inflow to acceptable levels. Grouting or other similar methods, as approved by the Engineer to suit the works condition shall be applied to minimise the groundwater infiltration.

Emergency Response Plan for Construction Site Discharges

5.7.34 The following measures should be implemented by the Contractors to minimise the chance of emergency construction site discharge (due to failure of treatment facilities such as sand traps, silt traps, sedimentation basins, oil interceptors etc.):

- Provide spare or standby treatment facilities of suitable capacities for emergency replacement in case damage or defect or malfunctioning of the duty treatment facilities is observed.

- Conduct daily integrity checking of the construction site drainage and treatment facilities to inspect malfunctions, in particular before, during and after a storm event.

- Carry out regular maintenance or desilting works to maintain effectiveness of the construction site drainage and treatment facilities in particular before, during and after a storm event.

5.7.35 An ERP should be developed to minimise the potential impact from construction site discharges under failure of treatment facilities during emergency situations or inclement weather. The ERP should give the emergency contacts to mobilise flood retention facilities and stakeholders to be notified as well as the details of the proposed construction site drainage system and the design and operation of duty and standby treatment facilities. The ERP should also provide the procedures and guidelines for routine integrity checking and maintenance of the drainage system and treatment facilities as well as the emergency response and rectification procedures to restore normal operation of the treatment facilities in case of treatment failure during emergency situation or inclement weather. The Best Management Practices (BMPs) in controlling water pollution arising from the construction activities and an event and action plan with action and limit levels for water quality monitoring should be included in the ERP. The ERP should be submitted to the EPD for approval before commencement of the construction works.

Operation Phase

Stormwater Management Practices and Stormwater Pollution Control Plan

5.7.36 The design of site drainage should follow the guidelines and practices as given in the ProPECC PN 1/23 "*Drainage Plan subject to Comments by the EPD*". With reference to Clause 2.3, Annex 6 of the EIAO-TM, mitigation measures including stormwater management practices to reduce stormwater pollution arising from the Project are as follows.

Design Measures to Control Erosion and Run-off Quantity

5.7.37 Exposed surface shall be avoided within the development sites to minimise soil erosion. The Project site shall be either hard paved or covered by landscaping area and plantation where appropriate.

5.7.38 The major rivers / streams / channels / nullahs within the Project site should be retained as far as practicable to maintain the original flow path. The drainage system should be designed to avoid flooding.

5.7.39 Green areas/tree/shrub planting etc. should be introduced within the Project site as far as possible including open space and along roadside amenity strips and central dividers, which can help to reduce soil erosion.

Devices / Facilities to Control Pollution

5.7.40 Screening facilities such as standard gully grating and trash grille, with spacing which is capable of screening large substances such as fallen leaves and rubbish should be provided at the inlet of drainage system.

5.7.41 Road gullies with standard design and silt traps and oil interceptors should be incorporated during the detailed design to remove particles present in stormwater run-off, where appropriate.

5.7.42 Evergreen tree species, which in general generate relatively smaller amount of fallen leaves, should be selected where possible.

Administrative Measures

5.7.43 Good management measures such as regular cleaning and sweeping of road surface/open areas is suggested. The road surface/open area cleaning should also be carried out prior to occurrence of rainstorm.

5.7.44 Manholes, as well as storm water gullies, ditches provided among the Project site should be regularly inspected and cleaned (e.g. monthly). Additional inspection and cleansing should be carried out before forecast heavy rainfall.

5.8 Cumulative Impacts

5.8.1 According to **Section 2**, the "*Development of Ma Tso Lung Area*" and site formation works of "*Development of Lok Ma Chau Loop*" would be constructed and operated concurrently with the Project.

Construction Phase

5.8.2 According to the Project Profile (No. PP-666/2024) of "*Development of Ma Tso Lung Area*", sources of potential water quality impacts during construction phase would include wastewater generated by general construction activities, construction site runoff, accidental spillage of chemicals, contaminated surface water and groundwater, sewage effluent from construction workforce, and runoff from removal/diversion of the existing watercourses. The potential water quality impacts could be mitigated and controlled by implementing the appropriate mitigation measures as outlined in the ProPECC PN 2/24 "*Construction Site Drainage*" and ETWB TC (Works) No. 5/2005 "*Protection of natural streams/rivers from adverse impacts arising from construction works*". No adverse water quality impact is anticipated.

5.8.3 According to the approved EIA report of "*Development of Lok Ma Chau Loop*", the potential water quality impacts associated with the site formation works include wastewater generated by general construction activities, construction site runoff, groundwater from contaminated area and sewage from workforce. With implementation of the recommended mitigation measures, no adverse water quality impact is anticipated.

5.8.4 As no significant water quality impacts are expected from the Project, the "*Development of Ma Tso Lung Area*" and "*Development of Lok Ma Chau Loop*" during construction phase, no adverse cumulative water quality impact is anticipated.

Operation Phase

5.8.5 According to the Project Profile (No. PP-666/2024) of "*Development of Ma Tso Lung Area*", proper drainage and sewerage systems would be provided in Ma Tso Lung development area to intercept surface run-off from the open paved areas and discharge into storm drains via adequately designed sand/silt removal facilities such as sand traps, silt traps and sedimentation basins. Silt removal facilities, channels and manholes would be maintained and the deposited silt and grit should be removed regularly, at the onset of and after each rainstorm. During emergency or maintenance condition of District Cooling System (DCS) plant, spent effluent would be discharged to the sewerage system. Precautionary measures will be incorporated in the design of the proposed sewage pumping station(s) and contingency plan will also be developed to avoid and minimise the emergency sewage bypass. With

implementation of the above proposed mitigation measures, no adverse water quality impact is anticipated.

5.8.6 According to the approved EIA report of "*Development of Lok Ma Chau Loop*", all the sewage generated from the LMC Loop would be collected and treated in the on-site STW. Precaution measures (such as standby units, dual power supply, etc.) would be included in the on-site STW design to minimise the occurrence of emergency sewage bypass. During emergency or maintenance condition of DCS plant, spent cooling waters would be discharged to the on-site STW. Adequate drainage system with silt traps and oil interceptors would be provided to collect the stormwater runoff. With implementation of the recommended mitigation measures, no adverse water quality impact is anticipated.

5.8.7 As neither the Project nor the concurrent projects are anticipated to generate significant water quality impact during operation phase, adverse cumulative impacts are hence not anticipated.

5.9 Evaluation of Residual Impacts

5.9.1 With proper implementation of mitigation measures, no adverse residual water quality impact is expected during construction and operation phases.

5.10 Environmental Monitoring and Audit

5.10.1 With full implementation of the mitigation measures such as good site practices, storm water pollution control plan, proper construction method, etc, no adverse impact is anticipated. No residual impact and cumulative impact are anticipated during both the construction and operation phases of the Project. In order to ensure effectiveness of the implemented mitigation measures, regular water quality monitoring in the LMC Meander is recommended during the construction phase. Details of the audit requirements are provided in the stand-alone EM&A Manual for the Project.

5.11 Conclusion

5.11.1 The potential water quality impacts associated with the land-based construction works include construction works in LMC Meander and EA, wastewater generated from general construction activities, construction site runoff, construction works near watercourses, diversion of watercourse, removal of ponds and wet areas, accidental spillage of chemicals, sewage from construction workforce, and groundwater infiltration arising from formation of underpass/depressed road. Given these impacts could be mitigated and controlled by implementing the recommended mitigation measures, no adverse water quality impact during construction phase is anticipated. Regular site inspections and water quality monitoring should be undertaken routinely to inspect the construction activities and works area to ensure the recommended mitigation measures are proper implemented.

5.11.2 The key issues of the operation phase are the surface runoff generated from the new paved areas. Provided that the recommended mitigation measures for the drainage system are properly implemented, the associated water quality impacts are expected to be minimal and acceptable. No adverse water quality impact during operation phase is therefore anticipated.

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